$(m-1)$-dimensional subspace of $\mathbb{R}^D$. Generally, the mth principal component can be found by subtracting the effect of the first $m-1$ principal components $\boldsymbol{b}_1, \dots, \boldsymbol{b}_{m-1}$ from the data, thereby trying to find principal components that compress the remaining information. We then arrive at the new data matrix

$$\hat{\boldsymbol{X}} := \boldsymbol{X} - \sum_{i=1}^{m-1} \boldsymbol{b}_i \boldsymbol{b}_i^\top \boldsymbol{X} = \boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X}\,, \tag{10.17}$$

where $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_N] \in \mathbb{R}^{D\times N}$ contains the data points as column vectors and $\boldsymbol{B}_{m-1} := \sum_{i=1}^{m-1} \boldsymbol{b}_i \boldsymbol{b}_i^\top$ is a projection matrix that projects onto the subspace spanned by $\boldsymbol{b}_1, \dots, \boldsymbol{b}_{m-1}$.

The matrix $\hat{\boldsymbol{X}} := [\hat{\boldsymbol{x}}_1, \dots, \hat{\boldsymbol{x}}_N] \in \mathbb{R}^{D\times N}$ in (10.17) contains the information in the data that has not yet been compressed.

Remark (Notation). Throughout this chapter, we do not follow the convention of collecting data $\boldsymbol{x}_1, \dots, \boldsymbol{x}_N$ as the rows of the data matrix, but we define them to be the columns of $\boldsymbol{X}$. This means that our data matrix $\boldsymbol{X}$ is a $D \times N$ matrix instead of the conventional $N \times D$ matrix. The reason for our choice is that the algebra operations work out smoothly without the need to either transpose the matrix or to redefine vectors as row vectors that are left-multiplied onto matrices. ♢

To find the mth principal component, we maximize the variance

$$V_m = \mathbb{V}[z_m] = \frac{1}{N}\sum_{n=1}^{N} z_{mn}^2 = \frac{1}{N}\sum_{n=1}^{N} (\boldsymbol{b}_m^\top \hat{\boldsymbol{x}_n})^2 = \boldsymbol{b}_m^\top \hat{\boldsymbol{S}} \boldsymbol{b}_m\,, \tag{10.18}$$

subject to $\|\boldsymbol{b}_m\|^2 = 1$, where we followed the same steps as in (10.9b) and defined $\hat{\boldsymbol{S}}$ as the data covariance matrix of the transformed dataset $\hat{\mathcal{X}} := \{\hat{\boldsymbol{x}}_1, \dots, \hat{\boldsymbol{x}}_N\}$. As previously, when we looked at the first principal component alone, we solve a constrained optimization problem and discover that the optimal solution $\boldsymbol{b}_m$ is the eigenvector of $\hat{\boldsymbol{S}}$ that is associated with the largest eigenvalue of $\hat{\boldsymbol{S}}$.

It turns out that $\boldsymbol{b}_m$ is also an eigenvector of $\boldsymbol{S}$. More generally, the sets of eigenvectors of $\boldsymbol{S}$ and $\hat{\boldsymbol{S}}$ are identical. Since both $\boldsymbol{S}$ and $\hat{\boldsymbol{S}}$ are symmetric, we can find an ONB of eigenvectors (spectral theorem 4.15), i.e., there exist D distinct eigenvectors for both $\boldsymbol{S}$ and $\hat{\boldsymbol{S}}$. Next, we show that every eigenvector of $\boldsymbol{S}$ is an eigenvector of $\hat{\boldsymbol{S}}$. Assume we have already found eigenvectors $\boldsymbol{b}_1, \dots, \boldsymbol{b}_{m-1}$ of $\hat{\boldsymbol{S}}$. Consider an eigenvector $\boldsymbol{b}_i$ of $\boldsymbol{S}$, i.e., $\boldsymbol{S}\boldsymbol{b}_i = \lambda_i \boldsymbol{b}_i$. In general,

$$\hat{\boldsymbol{S}}\boldsymbol{b}_i = \frac{1}{N}\hat{\boldsymbol{X}}\hat{\boldsymbol{X}}^\top \boldsymbol{b}_i = \frac{1}{N}(\boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X})(\boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X})^\top \boldsymbol{b}_i \tag{10.19a}$$

$$= (\boldsymbol{S} - \boldsymbol{S}\boldsymbol{B}_{m-1} - \boldsymbol{B}_{m-1}\boldsymbol{S} + \boldsymbol{B}_{m-1}\boldsymbol{S}\boldsymbol{B}_{m-1})\boldsymbol{b}_i\,. \tag{10.19b}$$

We distinguish between two cases. If $i \geqslant m$, i.e., $\boldsymbol{b}_i$ is an eigenvector that is not among the first $m-1$ principal components, then $\boldsymbol{b}_i$ is orthogonal to the first $m-1$ principal components and $\boldsymbol{B}_{m-1}\boldsymbol{b}_i = \boldsymbol{0}$. If $i < m$, i.e., $\boldsymbol{b}_i$ is among the first $m-1$ principal components, then $\boldsymbol{b}_i$ is a basis vector

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of the principal subspace onto which $\boldsymbol{B}_{m-1}$ projects. Since $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_{m-1}$ are an ONB of this principal subspace, we obtain $\boldsymbol{B}_{m-1}\boldsymbol{b}_i = \boldsymbol{b}_i$. The two cases can be summarized as follows:

$$\boldsymbol{B}_{m-1}\boldsymbol{b}_i = \boldsymbol{b}_i \quad \text{if } i < m\,, \qquad \boldsymbol{B}_{m-1}\boldsymbol{b}_i = \boldsymbol{0} \quad \text{if } i \geqslant m\,. \tag{10.20}$$

In the case $i \geqslant m$, by using (10.20) in (10.19b), we obtain $\hat{\boldsymbol{S}}\boldsymbol{b}_i = (\boldsymbol{S} - \boldsymbol{B}_{m-1}\boldsymbol{S})\boldsymbol{b}_i = \boldsymbol{S}\boldsymbol{b}_i = \lambda_i \boldsymbol{b}_i$, i.e., $\boldsymbol{b}_i$ is also an eigenvector of $\hat{\boldsymbol{S}}$ with eigenvalue λ_i. Specifically,

$$\hat{\boldsymbol{S}}\boldsymbol{b}_m = \boldsymbol{S}\boldsymbol{b}_m = \lambda_m \boldsymbol{b}_m\,. \tag{10.21}$$

Equation (10.21) reveals that $\boldsymbol{b}_m$ is not only an eigenvector of $\boldsymbol{S}$ but also of $\hat{\boldsymbol{S}}$. Specifically, λ_m is the largest eigenvalue of $\hat{\boldsymbol{S}}$ and λ_m is the mth largest eigenvalue of $\boldsymbol{S}$, and both have the associated eigenvector $\boldsymbol{b}_m$.

In the case $i < m$, by using (10.20) in (10.19b), we obtain

$$\hat{\boldsymbol{S}}\boldsymbol{b}_i = (\boldsymbol{S} - \boldsymbol{S}\boldsymbol{B}_{m-1} - \boldsymbol{B}_{m-1}\boldsymbol{S} + \boldsymbol{B}_{m-1}\boldsymbol{S}\boldsymbol{B}_{m-1})\boldsymbol{b}_i = \boldsymbol{0} = 0\boldsymbol{b}_i \tag{10.22}$$

This means that $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_{m-1}$ are also eigenvectors of $\hat{\boldsymbol{S}}$, but they are associated with eigenvalue 0 so that $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_{m-1}$ span the null space of $\hat{\boldsymbol{S}}$.

Overall, every eigenvector of $\boldsymbol{S}$ is also an eigenvector of $\hat{\boldsymbol{S}}$. However, if the eigenvectors of $\boldsymbol{S}$ are part of the $(m-1)$ dimensional principal subspace, then the associated eigenvalue of $\hat{\boldsymbol{S}}$ is 0.

This derivation shows that there is an intimate connection between the M-dimensional subspace with maximal variance and the eigenvalue decomposition. We will revisit this connection in Section 10.4.

With the relation (10.21) and $\boldsymbol{b}_m^\top \boldsymbol{b}_m = 1$, the variance of the data projected onto the mth principal component is

$$V_m = \boldsymbol{b}_m^\top \boldsymbol{S}\boldsymbol{b}_m \overset{(10.21)}{=} \lambda_m \boldsymbol{b}_m^\top \boldsymbol{b}_m = \lambda_m\,. \tag{10.23}$$

This means that the variance of the data, when projected onto an M-dimensional subspace, equals the sum of the eigenvalues that are associated with the corresponding eigenvectors of the data covariance matrix.

Example 10.2 (Eigenvalues of MNIST "8")

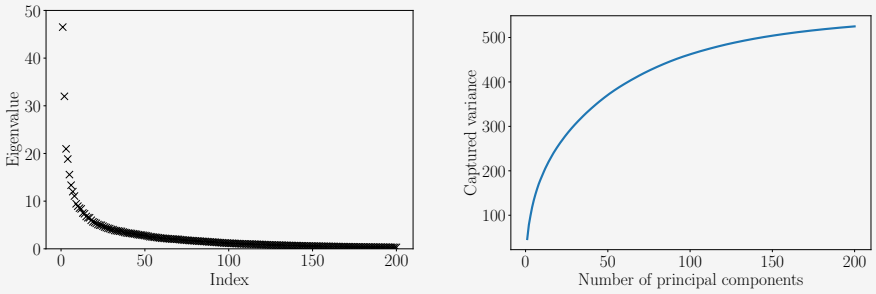


Figure 10.5 Properties of the training data of MNIST "8". (a) Eigenvalues sorted in descending order; (b) Variance captured by the principal components associated with the largest eigenvalues.

(a) Eigenvalues (sorted in descending order) of the data covariance matrix of all digits "8" in the MNIST training set.

(b) Variance captured by the principal components.

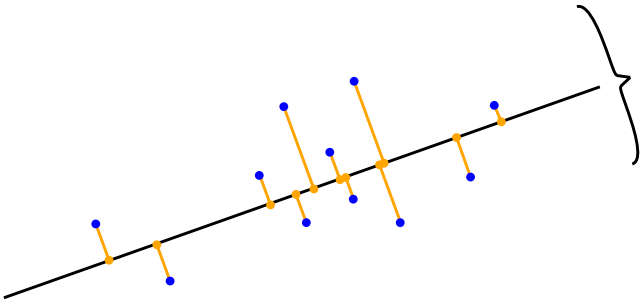


Figure 10.6 Illustration of the projection approach: Find a subspace (line) that minimizes the length of the difference vector between projected (orange) and original (blue) data.

Taking all digits "8" in the MNIST training data, we compute the eigenvalues of the data covariance matrix. Figure 10.5(a) shows the 200 largest eigenvalues of the data covariance matrix. We see that only a few of them have a value that differs significantly from 0. Therefore, most of the variance, when projecting data onto the subspace spanned by the corresponding eigenvectors, is captured by only a few principal components, as shown in Figure 10.5(b).

Overall, to find an M-dimensional subspace of $\mathbb{R}^D$ that retains as much information as possible, PCA tells us to choose the columns of the matrix $\boldsymbol{B}$ in (10.3) as the M eigenvectors of the data covariance matrix $\boldsymbol{S}$ that are associated with the M largest eigenvalues. The maximum amount of variance PCA can capture with the first M principal components is

$$V_M = \sum_{m=1}^{M} \lambda_m \,, \tag{10.24}$$

where the λ_m are the M largest eigenvalues of the data covariance matrix $\boldsymbol{S}$. Consequently, the variance lost by data compression via PCA is

$$J_M := \sum_{j=M+1}^{D} \lambda_j = V_D - V_M \,. \tag{10.25}$$

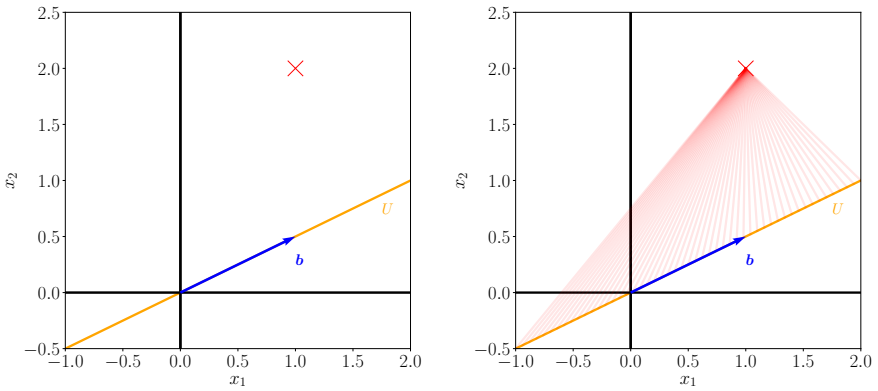
Instead of these absolute quantities, we can define the relative variance captured as $\frac{V_M}{V_D}$, and the relative variance lost by compression as $1 - \frac{V_M}{V_D}$.

10.3 Projection Perspective

In the following, we will derive PCA as an algorithm that directly minimizes the average reconstruction error. This perspective allows us to interpret PCA as implementing an optimal linear auto-encoder. We will draw heavily from Chapters 2 and 3.

In the previous section, we derived PCA by maximizing the variance in the projected space to retain as much information as possible. In the

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(a) Setting.

(b) Differences $\boldsymbol{x} - \tilde{\boldsymbol{x}}_i$ for 50 different $\tilde{\boldsymbol{x}}_i$ are shown by the red lines.

Figure 10.7 Simplified projection setting. (a) A vector $\boldsymbol{x} \in \mathbb{R}^2$ (red cross) shall be projected onto a one-dimensional subspace $U \subseteq \mathbb{R}^2$ spanned by $\boldsymbol{b}$. (b) shows the difference vectors between $\boldsymbol{x}$ and some candidates $\tilde{\boldsymbol{x}}$.

following, we will look at the difference vectors between the original data $\boldsymbol{x}_n$ and their reconstruction $\tilde{\boldsymbol{x}}_n$ and minimize this distance so that $\boldsymbol{x}_n$ and $\tilde{\boldsymbol{x}}_n$ are as close as possible. Figure 10.6 illustrates this setting.

10.3.1 Setting and Objective

Assume an (ordered) orthonormal basis (ONB) $B = (\boldsymbol{b}_1, \ldots, \boldsymbol{b}_D)$ of $\mathbb{R}^D$, i.e., $\boldsymbol{b}_i^\top \boldsymbol{b}_j = 1$ if and only if $i = j$ and 0 otherwise.

From Section 2.5 we know that for a basis $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_D)$ of $\mathbb{R}^D$ any $\boldsymbol{x} \in \mathbb{R}^D$ can be written as a linear combination of the basis vectors of $\mathbb{R}^D$, i.e.,

Vectors $\tilde{\boldsymbol{x}} \in U$ could be vectors on a plane in $\mathbb{R}^3$. The dimensionality of the plane is 2, but the vectors still have three coordinates with respect to the standard basis of $\mathbb{R}^3$.

$$\boldsymbol{x} = \sum_{d=1}^{D} \zeta_d \boldsymbol{b}_d = \sum_{m=1}^{M} \zeta_m \boldsymbol{b}_m + \sum_{j=M+1}^{D} \zeta_j \boldsymbol{b}_j \tag{10.26}$$

for suitable coordinates $\zeta_d \in \mathbb{R}$.

We are interested in finding vectors $\tilde{\boldsymbol{x}} \in \mathbb{R}^D$, which live in lower-dimensional subspace $U \subseteq \mathbb{R}^D$, $\dim(U) = M$, so that

$$\tilde{\boldsymbol{x}} = \sum_{m=1}^{M} z_m \boldsymbol{b}_m \in U \subseteq \mathbb{R}^D \tag{10.27}$$

is as similar to $\boldsymbol{x}$ as possible. Note that at this point we need to assume that the coordinates z_m of $\tilde{\boldsymbol{x}}$ and ζ_m of $\boldsymbol{x}$ are not identical.

In the following, we use exactly this kind of representation of $\tilde{\boldsymbol{x}}$ to find optimal coordinates $\boldsymbol{z}$ and basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M$ such that $\tilde{\boldsymbol{x}}$ is as similar to the original data point $\boldsymbol{x}$ as possible, i.e., we aim to minimize the (Euclidean) distance $\|\boldsymbol{x} - \tilde{\boldsymbol{x}}\|$. Figure 10.7 illustrates this setting.

Without loss of generality, we assume that the dataset $\mathcal{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$, $\boldsymbol{x}_n \in \mathbb{R}^D$, is centered at $\boldsymbol{0}$, i.e., $\mathbb{E}[\mathcal{X}] = \boldsymbol{0}$. Without the zero-mean assump-

tion, we would arrive at exactly the same solution, but the notation would be substantially more cluttered.

We are interested in finding the best linear projection of $\mathcal{X}$ onto a lower-dimensional subspace U of $\mathbb{R}^D$ with $\dim(U) = M$ and orthonormal basis vectors $\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M$. We will call this subspace U the *principal subspace*. The projections of the data points are denoted by

principal subspace

$$\tilde{\boldsymbol{x}}_n := \sum_{m=1}^{M} z_{mn}\boldsymbol{b}_m = \boldsymbol{B}\boldsymbol{z}_n \in \mathbb{R}^D\,, \tag{10.28}$$

where $\boldsymbol{z}_n := [z_{1n}, \ldots, z_{Mn}]^\top \in \mathbb{R}^M$ is the coordinate vector of $\tilde{\boldsymbol{x}}_n$ with respect to the basis $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$. More specifically, we are interested in having the $\tilde{\boldsymbol{x}}_n$ as similar to $\boldsymbol{x}_n$ as possible.

The similarity measure we use in the following is the squared distance (Euclidean norm) $\|\boldsymbol{x} - \tilde{\boldsymbol{x}}\|^2$ between $\boldsymbol{x}$ and $\tilde{\boldsymbol{x}}$. We therefore define our objective as minimizing the average squared Euclidean distance (*reconstruction error*) (Pearson, 1901)

reconstruction error

$$J_M := \frac{1}{N}\sum_{n=1}^{N} \|\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n\|^2\,, \tag{10.29}$$

where we make it explicit that the dimension of the subspace onto which we project the data is M. In order to find this optimal linear projection, we need to find the orthonormal basis of the principal subspace and the coordinates $\boldsymbol{z}_n \in \mathbb{R}^M$ of the projections with respect to this basis.

To find the coordinates $\boldsymbol{z}_n$ and the ONB of the principal subspace, we follow a two-step approach. First, we optimize the coordinates $\boldsymbol{z}_n$ for a given ONB $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$; second, we find the optimal ONB.

10.3.2 Finding Optimal Coordinates

Let us start by finding the optimal coordinates $z_{1n}, \ldots, z_{Mn}$ of the projections $\tilde{\boldsymbol{x}}_n$ for $n = 1, \ldots, N$. Consider Figure 10.7(b), where the principal subspace is spanned by a single vector $\boldsymbol{b}$. Geometrically speaking, finding the optimal coordinates z corresponds to finding the representation of the linear projection $\tilde{\boldsymbol{x}}$ with respect to $\boldsymbol{b}$ that minimizes the distance between $\tilde{\boldsymbol{x}} - \boldsymbol{x}$. From Figure 10.7(b), it is clear that this will be the orthogonal projection, and in the following we will show exactly this.

We assume an ONB $(\boldsymbol{b}_1, \ldots, \boldsymbol{b}_M)$ of $U \subseteq \mathbb{R}^D$. To find the optimal coordinates $\boldsymbol{z}_m$ with respect to this basis, we require the partial derivatives

$$\frac{\partial J_M}{\partial z_{in}} = \frac{\partial J_M}{\partial \tilde{\boldsymbol{x}}_n}\frac{\partial \tilde{\boldsymbol{x}}_n}{\partial z_{in}}\,, \tag{10.30a}$$

$$\frac{\partial J_M}{\partial \tilde{\boldsymbol{x}}_n} = -\frac{2}{N}(\boldsymbol{x}_n - \tilde{\boldsymbol{x}}_n)^\top \in \mathbb{R}^{1\times D}\,, \tag{10.30b}$$

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